

EOI Process Validation

Architecture & Data Model Documentation

Domain: Insurance / Group Benefits — Evidence of Insurability (EOI)

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This document describes the complete technical architecture, entity data model, field definitions, and process flows for the EOI Process Validation System.

1. Executive Summary

The **EOI (Evidence of Insurability) Process Validation System** is a rule-based automated underwriting platform designed to evaluate insurance coverage applications that exceed guaranteed issue limits. The system manages the complete lifecycle from application intake through validation, exception handling, underwriting decision, and audit compliance.

Key Capabilities:

Capability	Description
Automated Rule Validation	Configurable validation rules with priority and severity levels
Exception Management	Intelligent routing of failed validations to underwriter queues
Workflow Orchestration	State-machine driven application lifecycle tracking
Audit & Compliance	Comprehensive change logging for regulatory requirements
Hierarchical Data Model	Employer → Division → Class → Member → Coverage chain

2. Architecture Overview

The EOI architecture follows a layered design pattern with clear separation between intake, validation, underwriting, and compliance concerns. Each layer is independently scalable and replaceable.

Architecture Layers:

Layer	Components	Responsibility
Channel Layer	API, Web Portal, Paper, Bulk	Application intake from multiple sources
Orchestration Layer	App Service, Workflow Engine, Document Mgmt	Route, track, and manage application state
Validation Engine	Rule Registry, Execution Engine, Result Store	Evaluate business rules against application data
Exception & UW	Exception Queue, Underwriter Workbench, Decision Engine	Manual review and risk assessment
Audit & Compliance	Audit Log, Workflow History, Reports	Immutable logging and regulatory reporting
Data Persistence	Member, Plan, Application, Validation tables	Relational storage with referential integrity

Architecture Principles:

Principle	Description
Separation of Concerns	Validation logic is decoupled from workflow and data persistence
Configurability	Rules are data-driven, not hard-coded, enabling business user control
Auditability	Every state change and data mutation is logged immutably
Scalability	Async validation execution supports high-volume batch processing
Extensibility	Plugin architecture allows new rule categories without schema changes

3. System Components

3.1 Employer & Group Hierarchy

Manages organizational structure: Employer Group → Division → Class → Member. Determines eligibility, plan availability, and coverage rules.

3.2 Member Management

Central registry of insured members with demographics, employment status, and group hierarchy linkage. Tracks eligibility separately from enrollment.

3.3 Insurance Plan & Coverage

Defines available products (Life, Disability, Critical Illness) with coverage parameters. Member_Coverage links members to active plans.

3.4 EOI Application

Core transaction entity representing a request for coverage above guaranteed issue limits. Tracks status through the entire lifecycle.

3.5 Validation Engine

Rule Registry stores configurable rules; Execution Engine runs rules against data; Result Store persists outcomes for reporting and routing.

3.6 Exception & Underwriting

Failed validations create exceptions in a priority queue. Underwriters claim, review, and resolve exceptions through a workbench.

3.7 Workflow & Audit

State machine tracks application transitions. Immutable audit logs capture all data changes for HIPAA, SOX, and state insurance compliance.

4. Entity Relationship Model

The data model consists of 15 entities organized into 7 functional domains. All relationships enforce referential integrity through foreign key constraints.

Cardinality Summary:

Parent Entity	Child Entity	Card.	Relationship Description
Employer_Group	Group_Division	1:N	One group has many divisions
Group_Division	Group_Class	1:N	One division has many classes
Employer_Group	Member	1:N	One group has many members
Group_Division	Member	1:N	One division has many members (optional)
Group_Class	Member	1:N	One class has many members (optional)
Member	Member_Eligibility	1:N	One member has eligibility history
Member	Member_Coverage	1:N	One member has multiple coverages
Insurance_Plan	Member_Coverage	1:N	One plan covers many members
Member	EOI_Application	1:N	One member submits many applications
Insurance_Plan	EOI_Application	1:N	One plan has many applications
EOI_Application	Validation_Execution	1:N	One app has many validation runs
Validation_Execution	Validation_Result	1:N	One execution produces many results
Validation_Rule	Validation_Result	1:N	One rule appears in many results
EOI_Application	Exception_Queue	1:N	One app may have multiple exceptions
Underwriter	Exception_Queue	1:N	One underwriter handles many exceptions
EOI_Application	Workflow_History	1:N	One app has many status transitions
EOI_Application	Audit_Log	1:N	One app generates many audit entries

5. Field Descriptions

The following sections provide detailed field-level documentation for all 15 entities in the EOI Process Validation data model. Each table includes the field name, data type, nullability constraint, and business description.

5.1 Employer_Group

Field	Type	Null	Description
group_id	BIGINT	NO	Primary key. Unique identifier for the employer group.
group_number	VARCHAR(50)	NO	External reference number assigned by carrier or administrator. Unique.
group_name	VARCHAR(255)	NO	Legal or trading name of the employer organization.
employer_name	VARCHAR(255)	NO	Name of employing entity (may differ from group_name in trust arrangements).
group_type	VARCHAR(50)	NO	Classification: Corporate, Small Business, Association, Union, Government.
market_segment	VARCHAR(50)	YES	Market categorization: Commercial, Individual, Government, Voluntary.
status	VARCHAR(20)	NO	Lifecycle state: ACTIVE, INACTIVE, PENDING, TERMINATED.
effective_date	DATE	NO	Date when the group contract becomes effective.
termination_date	DATE	YES	Date when group contract ends. NULL if ongoing.
created_date	TIMESTAMP	NO	Row creation timestamp. Auto-populated.
updated_date	TIMESTAMP	NO	Last modification timestamp. Auto-updated.

5.2 Group_Division

Field	Type	Null	Description
division_id	BIGINT	NO	Primary key. Unique identifier for the division.
group_id	BIGINT	NO	Foreign key to Employer_Group. Establishes ownership.
division_code	VARCHAR(50)	NO	Short code for division (e.g., DIV-001, EAST_REGION).
division_name	VARCHAR(255)	NO	Descriptive name (e.g., Northeast Sales Division).
effective_date	DATE	NO	When division becomes active under group contract.
status	VARCHAR(20)	NO	ACTIVE, INACTIVE, or TERMINATED.

5.3 Group_Class

Field	Type	Null	Description
class_id	BIGINT	NO	Primary key. Unique identifier for the class.
division_id	BIGINT	NO	Foreign key to Group_Division. Classes belong to divisions.
class_code	VARCHAR(50)	NO	Short code (e.g., CLS-A, EXECUTIVE).
class_name	VARCHAR(255)	NO	Descriptive name (e.g., Executive Officers, Hourly Workers).
eligibility_rule	TEXT	YES	JSON or expression defining who qualifies (min hours, job grade).
status	VARCHAR(20)	NO	ACTIVE, INACTIVE, or TERMINATED.

5.4 Member

Field	Type	Null	Description
member_id	BIGINT	NO	Primary key. System-generated unique member identifier.
employee_number	VARCHAR(50)	NO	Employer's internal employee ID or badge number.
first_name	VARCHAR(100)	NO	Member's legal first name.
last_name	VARCHAR(100)	NO	Member's legal last name.
dob	DATE	NO	Date of Birth. Used for age-based eligibility and rate calculations.
gender	CHAR(1)	YES	M=Male, F=Female, O=Other, U=Unknown.
ssn	VARCHAR(11)	YES	Social Security Number. Must be encrypted at rest (AES-256).
email	VARCHAR(255)	YES	Primary email for notifications and portal access.
phone	VARCHAR(20)	YES	Primary contact phone number.
employment_status	VARCHAR(30)	NO	ACTIVE, TERMINATED, LEAVE, RETIRED. Drives eligibility.
hire_date	DATE	NO	Original date of hire. Used for waiting period calculations.
group_id	BIGINT	NO	Foreign key to Employer_Group. Required association.
division_id	BIGINT	YES	Foreign key to Group_Division. Optional sub-organization.
class_id	BIGINT	YES	Foreign key to Group_Class. Optional benefit class.
status	VARCHAR(20)	NO	Member record status: ACTIVE, INACTIVE, TERMINATED.

5.5 Member_Eligibility

Field	Type	Null	Description
eligibility_id	BIGINT	NO	Primary key. Unique eligibility determination record.
member_id	BIGINT	NO	Foreign key to Member. Links to the subject.
effective_date	DATE	NO	Start date of this eligibility period.
termination_date	DATE	YES	End date of eligibility. NULL if currently eligible.
eligibility_status	VARCHAR(30)	NO	ELIGIBLE, INELIGIBLE, PENDING, EXPIRED.
reason	TEXT	YES	Human-readable explanation for the determination.
verified_date	TIMESTAMP	YES	When eligibility was last verified by system or user.
verified_by	VARCHAR(100)	YES	User ID or system process that performed verification.
created_date	TIMESTAMP	NO	Record creation timestamp.
updated_date	TIMESTAMP	NO	Last update timestamp.

5.6 Insurance_Plan

Field	Type	Null	Description
plan_id	BIGINT	NO	Primary key. Unique plan identifier.
plan_code	VARCHAR(50)	NO	Carrier-specific plan code (e.g., GL-2026-BASIC). Unique.
plan_name	VARCHAR(255)	NO	Marketing name (e.g., Group Life Basic).
product_type	VARCHAR(50)	NO	Insurance line: Life, Disability, Critical Illness, Accident.
coverage_type	VARCHAR(50)	NO	Tier: Basic, Supplemental, Voluntary, Core.
guaranteed_issue	BOOLEAN	YES	True if no EOI required up to plan's GI limit.
maximum_coverage	DECIMAL(15,2)	YES	Absolute maximum coverage amount available.
minimum_coverage	DECIMAL(15,2)	YES	Minimum required enrollment amount. Default 0.
status	VARCHAR(20)	NO	ACTIVE, INACTIVE, TERMINATED.

5.7 Member_Coverage

Field	Type	Null	Description
coverage_id	BIGINT	NO	Primary key. Unique coverage enrollment record.
member_id	BIGINT	NO	Foreign key to Member. The insured individual.
plan_id	BIGINT	NO	Foreign key to Insurance_Plan. The enrolled product.
coverage_amount	DECIMAL(15,2)	NO	Dollar amount of coverage (e.g., \$150,000.00).
coverage_level	VARCHAR(50)	YES	Tier: Employee, Employee+Spouse, Employee+Child, Family.
coverage_status	VARCHAR(20)	NO	ACTIVE, PENDING, TERMINATED, SUSPENDED.
effective_date	DATE	NO	When coverage becomes active.
termination_date	DATE	YES	When coverage ends. NULL if active.

5.8 EOI_Application

Field	Type	Null	Description
application_id	BIGINT	NO	Primary key. System-generated application ID.
application_number	VARCHAR(50)	NO	Business reference (e.g., EOI-2026-0004521). Unique.
member_id	BIGINT	NO	Foreign key to Member. Applicant.
plan_id	BIGINT	NO	Foreign key to Insurance_Plan. Target plan.
requested_amount	DECIMAL(15,2)	NO	Additional coverage requested above current/GI limit.
application_date	DATE	NO	Date the application was submitted.
submission_channel	VARCHAR(50)	NO	Origin: PORTAL, PAPER, EMAIL, API, BULK_IMPORT.
application_status	VARCHAR(30)	NO	Business outcome: SUBMITTED, UNDER_REVIEW, APPROVED, DECLINED, PENDING_INFO, WITHDRAWN.
workflow_status	VARCHAR(30)	NO	Process stage: INTAKE, VALIDATION, UNDERWRITING, DECISION, CLOSED, EXCEPTION.
created_by	VARCHAR(100)	YES	User ID or system that created the application.
created_date	TIMESTAMP	NO	Submission timestamp.

5.9 Validation_Rule

Field	Type	Null	Description
rule_id	BIGINT	N0	Primary key. Unique rule identifier.
rule_name	VARCHAR(255)	N0	Human-readable name. Unique.
rule_category	VARCHAR(50)	N0	ELIGIBILITY, COVERAGE_LIMIT, DATA_QUALITY, COMPLIANCE, BUSINESS_RULE.
rule_expression	TEXT	N0	Validation logic: SQL, JSON Logic, or DSL reference.
priority	INT	N0	Execution order. Lower numbers run first (1=highest).
severity	VARCHAR(20)	N0	Impact: INFO, WARNING, ERROR, CRITICAL.
enabled	BOOLEAN	YES	If FALSE, rule is skipped during execution.
created_date	TIMESTAMP	N0	When the rule was defined.

5.10 Validation_Execution

Field	Type	Null	Description
execution_id	BIGINT	N0	Primary key. Unique validation run identifier.
application_id	BIGINT	N0	Foreign key to EOI_Application. Subject application.
started_time	TIMESTAMP	N0	When the validation run began.
completed_time	TIMESTAMP	YES	When validation finished. NULL if still running.
execution_status	VARCHAR(30)	N0	RUNNING, COMPLETED, FAILED, TIMEOUT.
rule_engine_version	VARCHAR(20)	YES	Semantic version of validation engine (e.g., 2.4.1).

5.11 Validation_Result

Field	Type	Null	Description
result_id	BIGINT	NO	Primary key. Unique result record.
execution_id	BIGINT	NO	Foreign key to Validation_Execution. Parent run.
rule_id	BIGINT	NO	Foreign key to Validation_Rule. The rule evaluated.
passed	BOOLEAN	NO	TRUE if application satisfied the rule; FALSE if failed.
expected_value	TEXT	YES	Value or condition the rule expected (for debugging).
actual_value	TEXT	YES	Actual value found in application data.
message	TEXT	YES	Human-readable explanation of result.
created_date	TIMESTAMP	NO	When result was recorded.

5.12 Underwriter

Field	Type	Null	Description
underwriter_id	BIGINT	NO	Primary key. Unique underwriter identifier.
employee_id	VARCHAR(50)	NO	HR system employee ID. Unique.
name	VARCHAR(255)	NO	Full name of the underwriter.
email	VARCHAR(255)	NO	Business email for notifications and SSO.
role	VARCHAR(50)	NO	JUNIOR, SENIOR, LEAD, MANAGER. Determines authority limits.
department	VARCHAR(100)	YES	Organizational unit (e.g., Group Life Underwriting).
status	VARCHAR(20)	NO	ACTIVE, INACTIVE, ON_LEAVE.

5.13 Exception_Queue

Field	Type	Null	Description
exception_id	BIGINT	NO	Primary key. Unique exception record.
application_id	BIGINT	NO	Foreign key to EOI_Application. Problematic application.
exception_type	VARCHAR(50)	NO	VALIDATION_FAILURE, MANUAL_REVIEW, SYSTEM_ERROR, POLICY_EXCEPTION.
priority	INT	NO	Triage: 1=Critical, 2=High, 3=Medium, 4=Low.
assigned_underwriter	BIGINT	YES	Foreign key to Underwriter. NULL if unassigned.
status	VARCHAR(20)	NO	OPEN, IN_PROGRESS, RESOLVED, ESCALATED.
created_date	TIMESTAMP	NO	When exception was generated.
resolved_date	TIMESTAMP	YES	When exception was closed. NULL until resolved.

5.14 Workflow_History

Field	Type	Null	Description
workflow_id	BIGINT	NO	Primary key. Unique transition record.
application_id	BIGINT	NO	Foreign key to EOI_Application. The subject.
previous_status	VARCHAR(30)	NO	Status before transition.
current_status	VARCHAR(30)	NO	Status after transition.
changed_by	VARCHAR(100)	NO	User ID or SYSTEM that triggered change.
changed_date	TIMESTAMP	NO	Exact timestamp of transition.
comments	TEXT	YES	Free-text explanation for status change.

5.15 Audit_Log

Field	Type	Null	Description
audit_id	BIGINT	N0	Primary key. Immutable audit record identifier.
application_id	BIGINT	YES	Foreign key to EOI_Application. Optional.
table_name	VARCHAR(100)	N0	Database table that was modified.
field_name	VARCHAR(100)	N0	Specific column that changed.
old_value	TEXT	YES	Previous value (NULL for inserts).
new_value	TEXT	YES	New value (NULL for deletes).
changed_by	VARCHAR(100)	N0	Identity of actor (user or system process).
changed_date	TIMESTAMP	N0	Timestamp of change. Immutable.

6. Process Flow

The EOI application follows a standardized lifecycle from initial submission through final closure. The workflow is state-machine driven with immutable history tracking.

6.1 Application Lifecycle Stages

Stage	Actor	Action	Outcome / Next State
INTAKE	Member/Admin	Submit EOI application with requested amount	Record created; workflow_status = INTAKE
PRE-VALIDATION	System	Check member eligibility and plan availability	If invalid, app declined immediately
RULE EXECUTION	Validation Engine	Execute all enabled rules by priority	Validation_Execution + Result records created
RESULT ANALYSIS	System	Aggregate results; check ERROR/CRITICAL failures	All pass → auto-approve or route to UW
EXCEPTION ROUTING	System	Create Exception_Queue entry for failures	Priority assigned by severity and amount
UW REVIEW	Underwriter	Claim exception, review, request info	Status updated to IN_PROGRESS
DECISION	Underwriter/System	Approve, decline, or pend for info	application_status updated; History recorded
COVERAGE UPDATE	System	If approved, update Member_Coverage	Member record synchronized
CLOSURE	System	Archive application; send notification	workflow_status = CLOSED

6.2 Decision Routing Logic

Auto-Approve Path: All rules pass AND requested_amount ≤ auto_approve_threshold → application_status = APPROVED; workflow_status = CLOSED.

Underwriting Path: All rules pass AND requested_amount > auto_approve_threshold → workflow_status = UNDERWRITING; assigned to underwriter by load balancing.

Exception Path: Any rule with severity ERROR or CRITICAL fails → workflow_status = EXCEPTION; Exception_Queue record created with priority = severity.

7. Validation Engine Design

The Validation Engine is the core decision-making component of the EOI system. It evaluates configurable business rules against application context data and produces structured results for routing and reporting.

7.1 Rule Categories

Category	Purpose	Example Rules
ELIGIBILITY	Verify member qualifies for EOI	Member is ACTIVE; hire date > 90 days ago
COVERAGE_LIMIT	Ensure amount within plan bounds	Requested <= Plan.maximum; Requested >= GI limit
DATA_QUALITY	Validate completeness and format	SSN is valid; Email present; DOB not future
COMPLIANCE	Regulatory and policy checks	Age at application < 70; State allows product
BUSINESS_RULE	Carrier-specific underwriting	No prior declinations in 12 months; Increase < 3x

7.2 Severity Levels

Severity	Behavior	Example
INFO	Logged only; no impact on routing	Member has multiple active coverages
WARNING	Logged; may trigger review if combined	Requested amount is 90% of maximum
ERROR	Fails validation; routes to exception queue	Member is not currently eligible
CRITICAL	Immediate decline; no underwriter review	Fraud indicator detected

7.3 Execution Pseudocode

Rules execute sequentially by priority (ascending). Execution continues through all rules unless a CRITICAL failure triggers an immediate hard stop.

```
FOR each enabled_rule IN rules ORDER BY priority ASC:
    result = evaluate(rule.expression, application_context)
    save_result(result)
    IF rule.severity == 'CRITICAL' AND result.passed == FALSE:
        mark_immediate_decline()
        BREAK
```

8. Integration Patterns

The EOI system integrates with upstream channels, downstream policy systems, and auxiliary services through well-defined interfaces.

8.1 Inbound Integrations

Source	Pattern	Data
Member Portal	REST API	Application submission, document upload
Employer HR System	SFTP / Batch	Member demographics, eligibility files
Underwriter Workbench	Web UI + API	Decisions, exception resolution
Document Management	Event-driven	Medical records, attestation forms

8.2 Outbound Integrations

Target	Pattern	Data
Member Portal	Webhook / Push	Status updates, decision notifications
Email Service	SMTP / API	Approval/decline letters, info requests
Policy Admin System	REST / SOAP	Approved coverage enrollments
Data Warehouse	ETL / Batch	Analytics, regulatory reporting
SIEM / Monitoring	Event Stream	Security events, anomaly detection

8.3 Domain Events

Key events published to the message bus for loose coupling:

- EOI.ApplicationSubmitted • EOI.ValidationStarted • EOI.ValidationCompleted
- EOI.ExceptionCreated • EOI.DecisionMade • EOI.CoverageActivated
- EOI.WorkflowStatusChanged

Appendix A: Data Retention & Compliance

Entity	Retention	Encryption	PII
Member	7 years post-termination	AES-256	Yes (SSN, DOB)
EOI_Application	10 years	AES-256	Yes
Audit_Log	10 years (immutable)	AES-256	Yes
Validation_Result	7 years	AES-256	No
Exception_Queue	7 years	AES-256	Yes

Appendix B: Glossary

Term	Definition
EOI	Evidence of Insurability — medical or financial proof required for coverage above standard limits
Guaranteed Issue (GI)	Coverage amount available without health questions or underwriting
Underwriter	Insurance professional who evaluates risk and makes coverage decisions
Exception Queue	Worklist of applications requiring manual review due to validation failures
Rule Engine	Software component that evaluates business rules against application data
Workflow Status	Technical process stage (Intake → Validation → Underwriting → Decision → Closed)
Application Status	Business outcome state (Submitted, Approved, Declined, etc.)

*Document generated from EOI_Validation_Process entity data model.
For questions, contact the Architecture Team.*